A Statistic Analysis of Reading Tendencies Among DVC Students

Theme: Investigating the differences between male and female DVC students regarding their favorite book genres and weekly reading hours for pleasure.

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Description of Populations and Samples

Populations
- Population 1: All female students in Diablo Valley College
- Population 2: All male students in Diablo Valley College

Samples
Sample 1: A sample of 33 female DVC students
Sample 2: A sample of 35 male DVC students

The two populations and their samples are independent. The selection of individual students form the female population has no relational, natural pairing, or matching effect on the selection of any student from the male population.

Description of Sampling Technique

For this project, I approached random students and asked about their weekly reading and genre preferences, successfully gathering 68 independent responses. While the survey was primarily centered in the campus cafeteria, I did not restrict to one fixed place. To increase randomness, I also collected data from outdoor pathways and library, approaching random students. This approach goes beyond a mere convenience sample because I did not rely on my close friends or a single online group chat. By stepping out around our campus and surveying anonymous individuals, I collected diverse samples representing various major and backgrounds.

Despite this efforts, two major biasses must be acknowledged:
- **Location Bias (Cafeteria Sampling)**: Although I collected some data from the library and walkways, the majority of our sample comes from the cafeteria. Students who read extensively for leisure are more likely to spend their free time in quiet zones like the library rather than the cafeteria. Therefore, the dataset might slightly underestimate the true average reading hours of the general DVC student population.
- **Selection and Non-Response Bias**: Due to the face-to-face nature of the survey in a social environment, I naturally tended to approach students who looked approachable or relaxed. Furthermore, students who were highly stressed or in a rush likely declined to participate, skewing the sample toward students with available free time.

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- Question 1: What is your gender?

[Male ( ) Female ( ) Other ( )]

- Question 2: What genre of books do you prefer?(Choose one)

[Romance ( ) Essay/Journalism ( ) Mystery/Thriller ( ) Sci-Fi/Fantasy ( ) Biography/History ( ) Self-Help/Personal Development ( ) Other ( )]

- Question 3: Excluding academic materials, how much time a week do you spend reading books for pleasure? (For example, 2.5, 0, 3.94 hr)

[(   ) hr]

The raw data set

Sample 1: A sample of 33 female DVC students

Gender	Preferred Genre	Weekly Reading Hours
0	Biography/History	2.75
0	Romance	3.5
0	Romance	3.5
0	Mystery/Thriller	1.2
0	Romance	5.8
0	Self-Help/Personal Development	1.4
0	Mystery/Thriller	2.75
0	Romance	0.2
0	Romance	4.1
0	Sci-Fi/Fantasy	1.8
0	Self-Help/Personal Development	3.2
0	Mystery/Thriller	7.25
0	Mystery/Thriller	0.01
0	Romance	5
0	Essay/Journalism	2.15
0	Mystery/Thriller	3.9
0	Romance	9
0	Self-Help/Personal Development	0.8
0	Sci-Fi/Fantasy	3.5
0	Romance	3.6
0	Mystery/Thriller	1.45
0	Essay/Journalism	0.3
0	Romance	6.2
0	Self-Help/Personal Development	3
0	Essay/Journalism	1.75
0	Mystery/Thriller	4.85

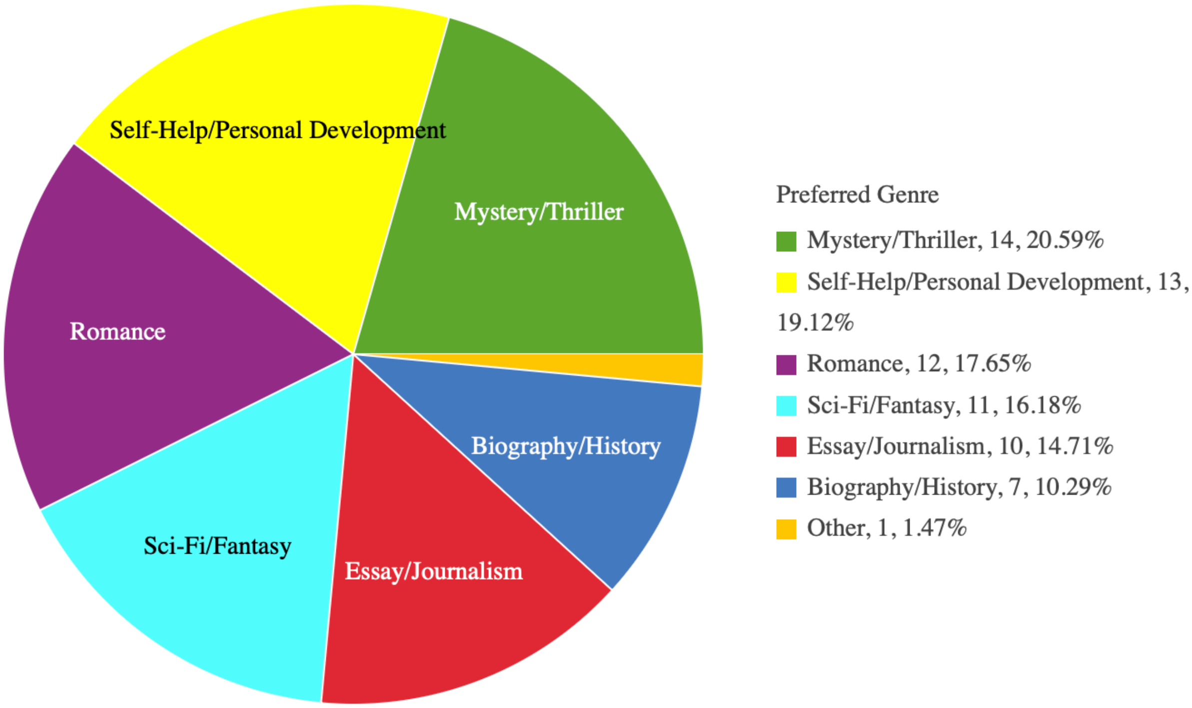
Gender	Preferred Genre	Weekly Reading Hours
0	Romance	2.4
0	Self-Help/Personal Development	0.5
0	Sci-Fi/Fantasy	8
0	Mystery/Thriller	3.1
0	Romance	5.5
0	Self-Help/Personal Development	0
0	Self-Help/Personal Development	2.9

Sample 2: A sample of 35 male DVC students

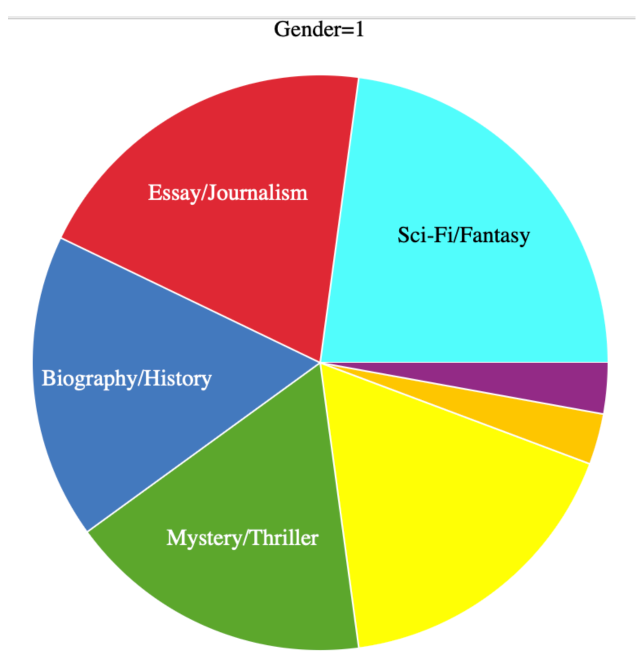
Gender	Preferred Genre	Weekly Reading Hours
1	Essay/Journalism	3
1	Self-Help/Personal Development	3.5
1	Sci-Fi/Fantasy	2
1	Mystery/Thriller	0.1
1	Essay/Journalism	3
1	Other	0.75
1	Mystery/Thriller	1
1	Biography/History	0.5
1	Sci-Fi/Fantasy	1.5
1	Biography/History	0
1	Biography/History	4.25
1	Self-Help/Personal Development	0.8
1	Essay/Journalism	2.1
1	Sci-Fi/Fantasy	5.5
1	Sci-Fi/Fantasy	0
1	Mystery/Thriller	1.2
1	Essay/Journalism	3
1	Self-Help/Personal Development	0.5

Gender	Preferred Genre	Weekly Reading Hours
1	Sci-Fi/Fantasy	1.75
1	Self-Help/Personal Development	0
1	Biography/History	7.4
1	Self-Help/Personal Development	2.25
1	Romance	1.1
1	Mystery/Thriller	0
1	Essay/Journalism	3.65
1	Sci-Fi/Fantasy	1.3
1	Self-Help/Personal Development	0.75
1	Biography/History	0
1	Mystery/Thriller	4.8
1	Sci-Fi/Fantasy	0.6
1	Essay/Journalism	0.11
1	Biography/History	6.15
1	Sci-Fi/Fantasy	1.85
1	Essay/Journalism	0.4
1	Mystery/Thriller	1.5

Overall Book Genre Preferences



The pie chart demonstrates that the overall distribution of reading preferences among DVC students is relatively evenly distributed across most major genres. Categories such as Mystery/Thriller, Self-Help/Personal Development, Romance, Sci-Fi/Fantasy, and Essay/Journalism show very little difference in frequency, suggesting a diverse and balanced interest in both fiction and practical non-fiction on campus. The only noticeable exception is Biography/History, which shows a distinct drop-off in responses. This indicates that while students are open to a wide variety of genres for leisure reading, they have a clear tendency to avoid traditional historical or biographical accounts during their free time

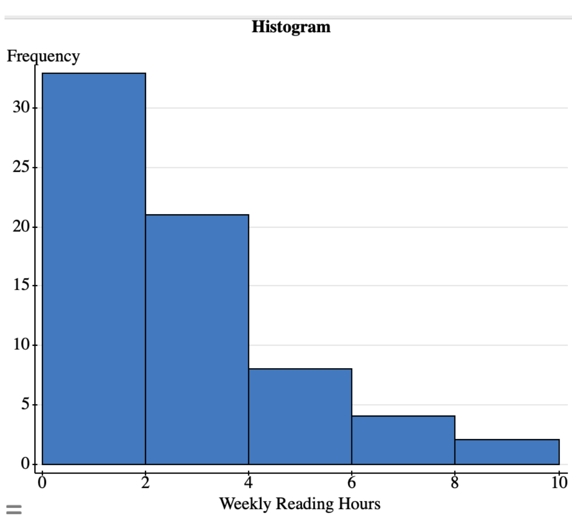


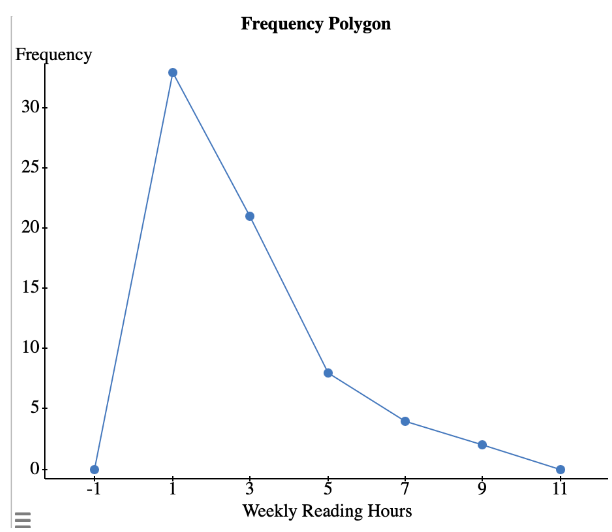
However, when the data is grouped by gender, a clear and meaningful difference in reading preferences emerges between female (0) and male (1) students. For female students, Romance (33.33%) stands out as the single most dominant preference. On the other hand, male students show a strong inclination toward Sci-Fi/Fantasy (22.86%), making it their top-rated genre. While the overall data showed a relatively uniform distribution, the gender-specific breakdown reveals literary preferences.

Weekly Reading Hours Frequency Distribution Table

Weekly Reading Hours	Frequency	Percentage Freqyency	Cumulative Frequency
0.00 - 1.99	33	48.53%	33
2.00 - 3.99	21	30.88%	54
4.00 - 5.99	8	11.76%	62
6.00 - 7.99	4	5.88%	66
8.00 - 10.00	2	2.94%	68

The frequency distribution table for Weekly Reading Hours, structured with a class width of 2 hours, reveals a textbook example of a strongly right-skewed distribution among DVC students. Specifically, 48.53 percent of students read for less than 2 hours per week. Furthermore, the cumulative frequency highlights that 54 out of 68 students (79.41%) engage in fewer than 4 hours of leisure reading weekly. As the weekly reading hours increase, the frequencies have a sharp and steady decline, dropping to 2.94% in the highest class. This frequency distribution table shows that a only 2.94% of the respondents are intensive readers who dedicate 8 to 10 hours a week to reading.





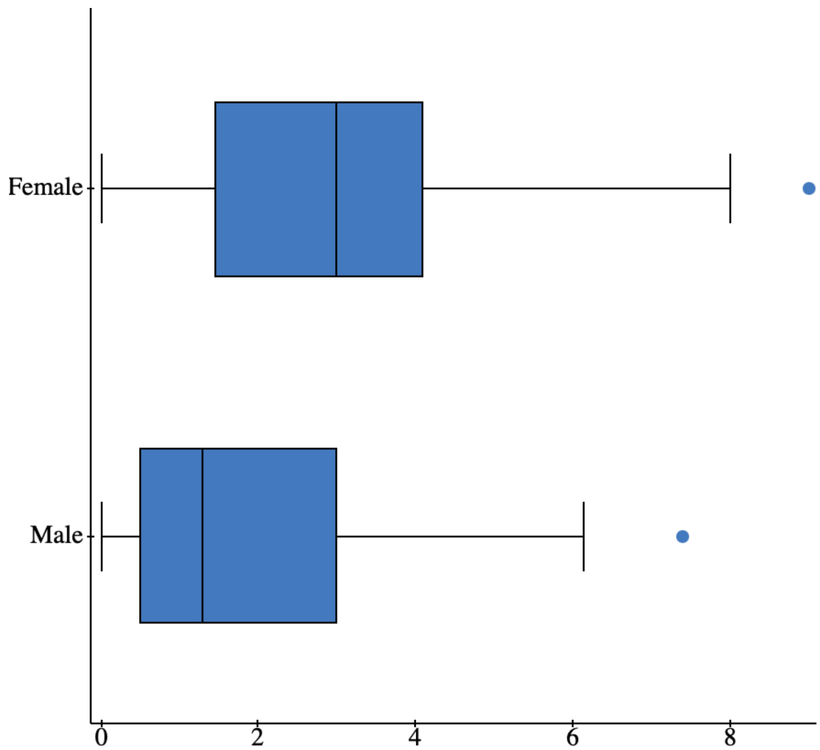
Both the histogram and the frequency polygon visually confirm that the distribution of Weekly Reading Hours among DVC students is strongly right-skewed. The histogram shows the tallest peak in the first class, with 33 respondents, creating a prominent left-heavy shape. As the reading hours increase, the height of the bars drops significantly. The line on the polygon reaches its lowest point in the upper classes of Reading Hours, emphasizing that very few students read beyond 6 hours a week.

Summary statistics for both independent samples using your quantitative data.

	Female	Male
Sample Size	33	35
Sample Mean	3.193	1.895
Sample Standard Deviation	2.3	1.906
Sample variance	5.292	3.632
5-Number Summary	(0, 1.45, 3, 4.1 ,9)	(0, 0.5, 1.3, 3, 7.4)
Range of usual values	0 - 7.793	0 - 5.707

The summary statistics provide a clear quantitative comparison between the weekly reading hours of female and male DVC students, revealing a distinct higher engagement among female respondents. On average, female students spend significantly more time reading for pleasure, with a sample mean of 3.19 hours compared to the male students' mean of 1.89 hours. Interestingly, the female sample variance of 5.29 and standard deviation of 2.30 hours are larger than the male variance of 3.63 and standard deviation of 1.91 hours, showing that female reading times are more spread out and diverse. Additionally, in the 5-number summary, the first quartile shows that while bottom 25% of female students read at least 1.45 hours or more, the bottom 25% of male students read 0.50 hours or less, indicating a much lower baseline of reading interest among males. Lastly, the range of usual values for male students, extending from 0 to 5.71 per week, is narrower that female students’ range, where between 0 and 7.79.

Two modified box plots, one for each 5-number summary



Column	Min	Q1	Median	Q3	Max
Female	0	1.45	3	4.1	9
Male	0	0.5	1.3	3	7.4

The box plots show that female students generally read much more than male students. The female box sits much higher on the graph, meaning their middle reading time (3.0 hours) is more than double the male middle time (1.3 hours). In addition, even the Q1 point of female readers (1.45 hours) beats the middle male reader. Finally, looking at the highest reading time of 7.4 hours among male students, it is a normal top score for females, but for males, it is so unusually high that the graph marks it as an outlier, showing that this specific male student is a rare exception.

Compare two populations with a 95% confidence interval

95% confidence interval results:

Difference	Sample Diff.	Std. Err.	DF	L. Limit	U. Limit
$\mu_1 - \mu_2$	1.2981558	0.51394393	62.270985	0.2708855	2.3254262

We are 95% confident that the interval from 0.271 to 2.325 actually does contain the true value of the population mean difference $\mu 1 - \mu 2$. This is a short and acceptable way of saying that if we were to select many different random samples of size 68 and construct the corresponding confidence intervals, 95% of them would contain the true difference between the mean weekly leisure reading hours of female and male students at DVC. In this correct interpretation, the confidence level of 95% refers to the success rate of the process used to estimate the population mean difference.

Test of hypothesis to compare your two populations

Claim: The mean weekly reading hours for pleasure of female DVC students is greater than the mean weekly reading hours for pleasure of male DVC students.

Hypothesis

Null Hypothesis: $\mu 1 = \mu 2$ (There is no difference between the mean weekly reading hours for pleasure of female and male DVC students.)

Alternative Hypothesis: $\mu 1 > \mu 2$ (The mean weekly reading hours for pleasure of female DVC students is greater than the mean weekly reading hours for pleasure of male DVC students.)

Hypothesis test results:

Difference	Sample Diff.	Std. Err.	DF	T-Stat	P-value
$\mu_1 - \mu_2$	1.2981558	0.51394393	62.270985	2.5258706	0.007

Since the p-value (0.007) is less than the significance level of 0.05, we reject the null hypothesis. There is sufficient evidence to support the claim that the mean weekly reading hours for pleasure of female students is greater than that of male students at DVC. The data conclusively shows that female students, on average, spend more time reading for pleasure than male students, and the observed sample difference of 1.30 hours is statistically significant and not due to random chance.

Conclusion

In conclusion, combining the sample data with confidence intervals and hypothesis testing allows us to make a valid conclusion about all students at DVC. While early summary stats and box plots showed that the female students read more, we cannot just assume this is true for the whole campus based on those numbers alone. However, statistical analysis gives us the proof we need to apply these findings to the entire DVC student body.

Based on the calculations, we were 95% confident that the difference in weekly reading hours for pleasure between all female and male DVC students is between 0.271 and 2.325 hours. Because this interval contains only positive numbers and does not include zero, it proves that female students overall read more than male students. This is supported by the hypothesis test, where very low p-value of 0.007 allows to reject the null hypothesis at the 0.05 confidence level. Together, these results prove that the 1.30 hour reading gap is statistically meaningful and due to real differences between the two groups, not just random luck in the sample. Therefore, we can confidently conclude that female DVC students, as a whole population, spend significantly more time reading for pleasure than male DVC students.